

Open-Vocabulary Semantic Segmentation with Mask-adapted CLIP

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 arXiv

Video

Code

Demo



Where is **Saturn V**, blossom?



Where is **Oculus**, Ukulele?



Where is **Golden gate**, yacht?



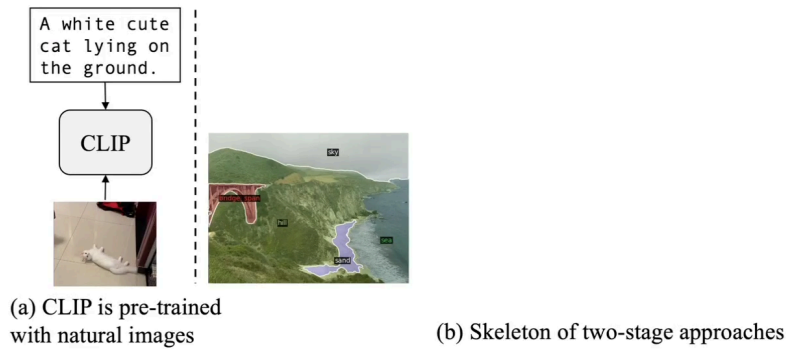
Our model can perform open-vocabulary segmentation with **user-defined arbitrary queries**.

Abstract

Open-vocabulary semantic segmentation aims to segment an image into semantic regions according to text descriptions, which may not have been seen during training. Recent two-stage methods first generate class-agnostic mask proposals and then leverage pre-trained vision-language models, *e.g.*, CLIP, to classify masked regions. We identify the performance bottleneck of this paradigm to be the pre-trained CLIP model, since it does not perform well on masked images. To address this, we propose to finetune CLIP on a collection of masked image regions and their corresponding text descriptions. We collect training data by mining an existing image-caption dataset (*e.g.*, COCO Captions), using CLIP to match masked image regions to nouns in the image captions. Compared with the more precise and manually annotated segmentation labels with fixed classes (*e.g.*, COCO-Stuff), we find our noisy but diverse dataset can better retain CLIP's generalization ability. Along with finetuning the entire model, we utilize the "blank" areas in masked images using a method we dub **mask prompt tuning**. Experiments demonstrate mask prompt tuning brings significant improvement without modifying any weights of CLIP, and it can further improve a fully finetuned model. In particular, when trained on COCO and evaluated on ADE20K-150, our best model achieves 29.6% mIoU, which is +8.5% higher than the previous state-of-the-art. **For the first time, open-vocabulary generalist models match the performance of supervised specialist models in 2017 without dataset specific adaptations.**

Motivation

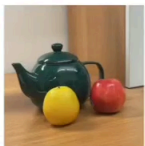
Our analysis reveals the pre-trained CLIP does **not** perform well on mask proposals, making it the performance bottleneck of two-stage approaches.



- (a) CLIP is pre-trained with natural images with little data augmentation.
- (b) Two-stage open-vocabulary semantic segmentation approaches first generate class-agnostic mask proposals and then leverage pre-trained CLIP to do open-vocabulary classification. The input of the CLIP model is cropped masked images, which have huge domain gap from the natural images.
- (c) Our analysis reveals that pre-trained CLIP does not work well on masked images.

Method

Our model consists of one segmentation model, *e.g.*, MaskFormer, and one CLIP model.



We first train the modified MaskFormer as the open-vocabulary segmentation baseline (Section 3.1). Then we collect diverse mask-category pairs from image captions (Section 3.2) and adapt CLIP for masked images (Section 3.3).

Results

For the first time, we show open-vocabulary **generalist** models can match the performance of supervised **specialist** models without dataset specific adaptations.

method	backbone	training dataset	A-847	PC-459	A-150	PC-59	PAS-20
<i>Open-vocabulary generalist models.</i>							
SPNet	R-101	PASCAL-15	-	-	-	24.3	18.3
ZS3Net	R-101	PASCAL-15	-	-	-	19.4	38.3
LSeg	R-101	PASCAL-15	-	-	-	-	47.4
LSeg+	R-101	COCO Panoptic	2.5	5.2	13.0	36.0	59.0
SimBaseline	R-101c	COCO-Stuff-156	-	-	15.3	-	74.5
ZegFormer	R-50	COCO-Stuff-156	-	-	16.4	-	80.7
OpenSeg	R-101	COCO Panoptic	4.0	6.5	15.3	36.9	60.0
OVSeg (Ours)	R-101c	COCO-Stuff-171	7.1	11.0	24.8	53.3	92.6
LSeg+	Eff-B7	COCO Panoptic	3.8	7.8	18.0	46.5	-
OpenSeg	Eff-B7	COCO Panoptic	6.3	9.0	21.1	42.1	-
OVSeg (Ours)	Swin-B	COCO-Stuff-171	9.0	12.4	29.6	55.7	94.5
<i>Supervised specialist models.</i>							
FCN	FCN-8s	Same as test	-	-	29.4	37.8	-
Deeplab	R-101	Same as test	-	-	-	45.7	77.7
SelfTrain	Eff-L2	Same as test	-	-	-	-	90.0

BibTeX

```
@inproceedings{liang2023open,  
  title={Open-vocabulary semantic segmentation with mask-adapted clip},  
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  booktitle={Proceedings of the IEEE/CVF Conference on Computer Vision and Pa  
  pages={7061--7070},  
  year={2023}  
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